



KOA CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & APP

Installation

```
npm install koa koa-router koa-bodyparser
import Koa from 'koa'
const app = new Koa()
app.keys = ['secret-key'] // for signed cook...
app.use(bodyParser())
app.listen(3000)
```

04. ROUTING (KOA-ROUTER)

Workflow

```
import Router from '@koa/router'
const router = new Router({ prefix: '/api' })
router.get('/users', async ctx => {
  ctx.body = await User.findAll()
})
router.post('/users', async ctx => {
  const user = await User.create(ctx.request.bo...
  ctx.status = 201
  ctx.body = user
})
app.use(router.routes()).use(router.allowedMe...
```

02. MIDDLEWARE

Architecture

```
app.use(async (ctx, next) => {
  const start = Date.now()
  await next() // call downstream middleware
  const ms = Date.now() - start
  ctx.set('X-Response-Time', `${ms}ms`)
})
```

Onion model: middleware wraps around each other

05. ASYNC/AWAIT FLOW

CRUD API

```
app.use(async (ctx, next) => {
  try {
    await next()
  } catch (err) {
    ctx.status = err.status || 500
    ctx.body = { error: err.message }
    ctx.app.emit('error', err, ctx)
  }
})
```

Error middleware must be first to catch downstream errors

03. CONTEXT OBJECT

YAML / Config

```
ctx.request ctx.response // Koa's enhanced ...
ctx.req ctx.res // Node native
ctx.state.user // shared state ac...
ctx.session.userId // (koa-session)
ctx.app // app instance
ctx.body = {} // sets response b...
ctx.status = 200 // HTTP status code
```

06. ERROR HANDLING

Integration

```
app.on('error', (err, ctx) => {
  console.error('Server Error', err, ctx)
})
// Throw in handler:
const { createError } = require('http-errors')
throw createError(404, 'User not found')
throw createError(400, 'Invalid input')
ctx.throw(404, 'Not found') // shorthand
```



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. BODY PARSING

Hardening

```
import bodyParser from 'koa-bodyparser'
app.use(bodyParser({
  enableTypes: ['json', 'form'],
  jsonLimit: '5mb',
}))
// Access parsed body:
ctx.request.body // JSON or form data
// File uploads: koa-multer
```

10. COMMON MIDDLEWARE PACKAGES

Unit Tests

@koa/router	routing
koa-bodyparser	JSON/form parsing
koa-static	static file serving
koa-session	session management
koa-compress	gzip compression
koa-logger	request logging
koa-helmet	security headers
koa-jwt	JWT authentication

08. STATIC FILES

Observability

```
import serve from 'koa-static'
app.use(serve('./public', {
  maxAge: 86400, // 1 day cache
  gzip: true,
}))
import mount from 'koa-mount'
app.use(mount('/assets', serve('./dist')))
```

11. COMMON GOTCHAS

Commands

Error catching middleware must be registered FIRST
await next() is async forgetting await breaks the chain
ctx.body must be set or Koa returns 404 by default
ctx.request.body only set after bodyparser middleware
app.keys required for signed cookies or sessions

09. COOKIES & SESSIONS

Optimization

```
ctx.cookies.get('session') // unsigned
ctx.cookies.set('token', value, { httpOnly: t...
// Signed cookies (requires app.keys):
ctx.cookies.get('session', { signed: true })
// Session middleware:
import session from 'koa-session'
app.use(session({ key: 'sess', maxAge: 86400 ...
```